CSP-203: Software Tools & Technologies Lab

Chapter – 3

Document Processing: LaTeX (Lecture – 3)

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Beamer Outline

- Beamer Introduction
- Beamer – Title – Author – Date
- Section
- Subsection
- Content in slide
- Customizing themes
- Graphics and tables – drawing figures
- Mathematical Equations
- Bibliography
- Animations – Slide transitions

What is Beamer?

- Beamer is a flexible LATEX class for making slides and presentations
- LaTeX class for creating presentations → slides, seminar, prosper, powerdot
- Each Beamer document class is made up of a series of *frames*. Each frame produces one or more slides, depending on the slide's overlays

- GitHub - josephwright/beamer: A LaTeX class for producing presentations and slides

LaTeX Beamer class



LaTeX Beamer presentation screenshot

Developer(s)	Till Tantau, Joseph Wright, Vedran Miletić
Initial release	March 2003; 20 years ago
Stable release	3.63 / May 26, 2021; 2 years ago
Repository	github.com/josephwright/beamer
Written in	LaTeX, TeX
Operating system	Unix-like, Windows
Platform	TeX Live, MiKTeX

Beamer: Syntax

```
\documentclass [pdf] {beamer}
      \title{CSP203}
      \subtitle{STTL}
      \author{IITBhilai}

\begin{document}

%% Title frame
\begin{frame}
      \titlepage
\end{frame}

\end{document}
```



Beamer: Syntax

```
\documentclass [pdf] {beamer}
\usetheme{Madrid}
      \title{CSP203}
      \subtitle{STTL}
      \author{IITBhilai}

\begin{document}

%% Title frame
\begin{frame}
      \titlepage
\end{frame}

\end{document}
```



Beamer: Syntax (1)

Header Commands

```
\documentclass [pdf] {beamer}
    \usepackage[utf8]{inputenc}
    \usetheme{default}
    \usecolortheme{default}

    \title [CSP203] {STTL}
    \subtitle {L7}
    \author [AB] {Anand Baswade}
\begin{document}
\begin{frame}
    \titlepage
\end{frame}
\end{document}
```



\setbeamertemplate{navigation symbols} {}

\usetheme{Madrid}

- Default slides will come with a ***navigation bar*** at the bottom right.
- It allows users to move one slide forward or backward, move to the next section, etc.
- To eliminate this → \beamertemplatenavigationsymbolsempty

Beamer: Frames

Each beamer project is made up of a series of ***frames.***
Each ***frame*** produces one or more slides.

A Basic Frame

```
\begin {frame} [<option>]
      \frametitle {<frame title>}
       <frame body>
\end {frame}
```

```
\begin{frame}{First Slide}
 This is CSP203 course.
\end{frame}
```

Options:
- plain → No headlines, foot lines, sidebars
- b → Bottom alignment
- c → Center alignment
- t → Top alignment
- shrink = *n* → Shrink everything by *n* percent

Beamer: Themes

- For the appearance of the presentation
- Beamer classifies the ***Themes*** into five categories

Theme Category	Description	Default / Optional
Presentation Themes	Slide template	Default
Color Themes	Color scheme of slide template	Optional
Font Themes	Defines the fonts	Optional
Inner Themes	Defines inside of slide like bullets, boxes, etc.,	Optional
Outer Themes	Defines outside of slide head- and foot lines.	Optional

- Presentation Themes

Syntax: \usetheme{themename}

AnnArbor Antibes Bergen Berkeley Berlin Boadilla boxes CambridgeUS
Copenhagen Darmstadt default Dresden Frankfurt Goettingen Hannover Ilmenau
JuanLesPins Luebeck Madrid Malmoe Marburg Montpellier PaloAlto Pittsburgh
Rochester Singapore Szeged Warsaw

Beamer: Themes (1)

- Color Themes

Syntax: \usecolortheme {color theme name}

Complete : albatross, beetle, crane, dove, fly, seagull, wolverine, beaver
Inner : lily, orchid, rose
Outer : whale, seahorse, dolphin

- Inner Themes

Syntax: \useinnertheme {inner theme name}

circles , default, inmargin, rectangles, rounded

- Outer Themes

Syntax: \useoutertheme{outer theme name}

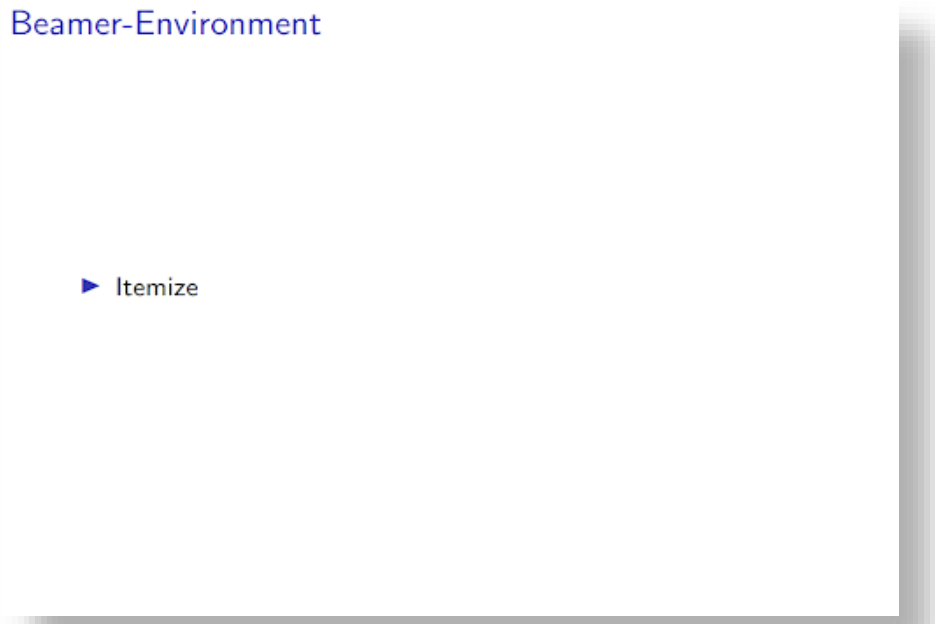
default	infolines	miniframes	sidebar	smoothbars
smoothtree	split tree			

Beamer: Environment - Lists

- Itemize

```
\begin{itemize}
\item <item name>
\end{itemize}
```

```
%% Beamer-Environment
\begin{frame}{Beamer-Environment}
        \begin{itemize}
            \item Itemize
        \end{itemize}
\end{frame}
```



- Enumerate

```
\begin{enumerate}
\item <item name>
\end{enumerate}
```

```
%% Beamer-Environment
\begin{frame}{Beamer-Environment}
        \begin{enumerate}
            \item Enumerate
        \end{enumerate}
\end{frame}
```



Beamer: Environment – Lists (1)

- Itemize

```
\begin{description}
\item <item name>
\end{description}
```

```
%% Beamer-Environment
\begin{frame}{Beamer-Environment}
        \begin{description}
            \item Itemize
        \end{description}
\end{frame}
```



Beamer: Environment – Mathematics Blocks

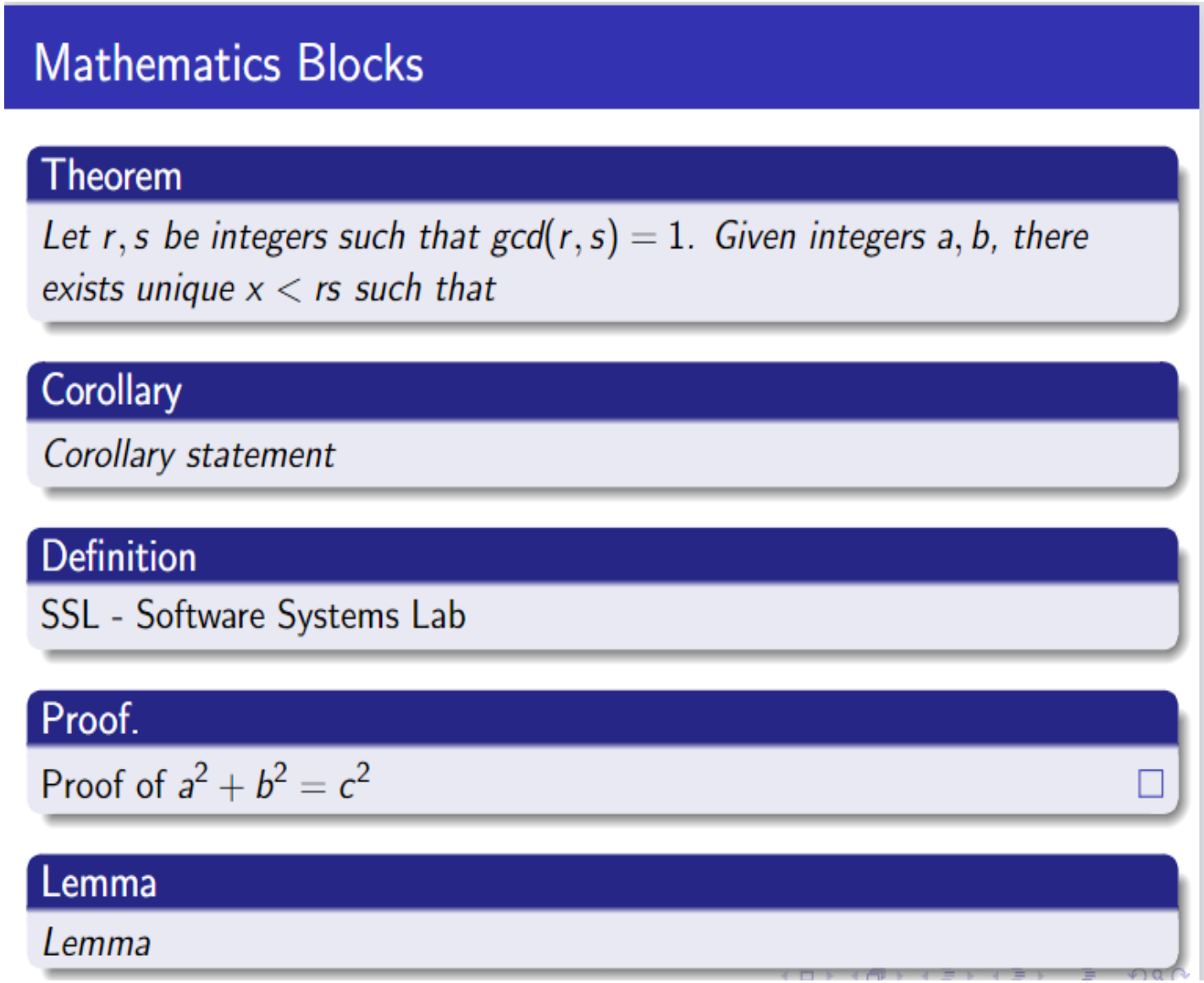
```
\begin{theorem}
Let $r, $s be integers such that gcd$(r, s)=1$.
Given integers $a,b$, there exists unique $x <rs$
such that
\end{theorem}
```

```
\begin{corollary}
Corollary statement
\end{corollary}
```

```
\begin{definition}
SSL - Software Systems Lab
\end{definition}
```

```
\begin{proof}
Proof of $a^2 + b^2 = c^2$
\end{proof}
```

```
\begin{lemma}
Lemma
\end{lemma}
```



Beamer: Animations

- Animation → The process of creating an illusion of motion or movement and rapidly showing a sequence of static figures that are (slightly) different.

- Uses the following packages:
  - animate
  - tikz

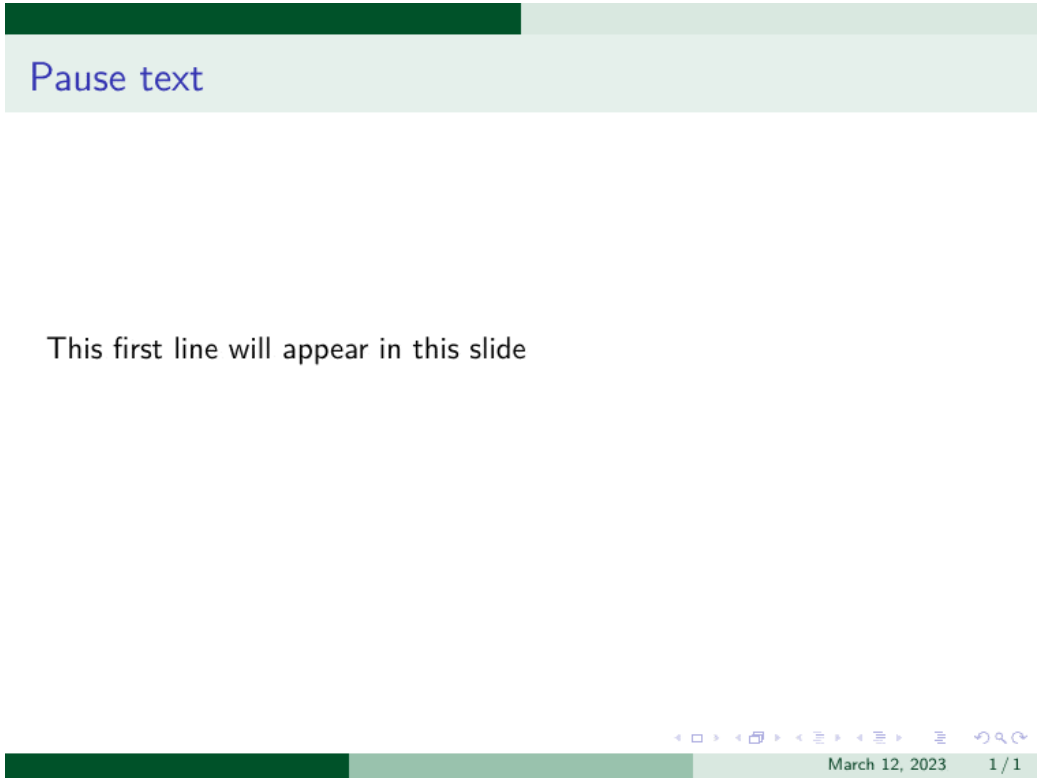
- Animations can be included for the following file formats
  - Static image or PNGs
  - Text
  - Videos or GIFs

Beamer: Animations - Text

When we have multiple lines of text, we can reveal these texts line by line using the ***pause*** command.

```
%% Beamer-Environment
\begin{frame}{Pause text}
        This first line will appear in this slide \pause
        the second text will be partially visible \pause
        And this last text will then appear
\end{frame}
```

Output:



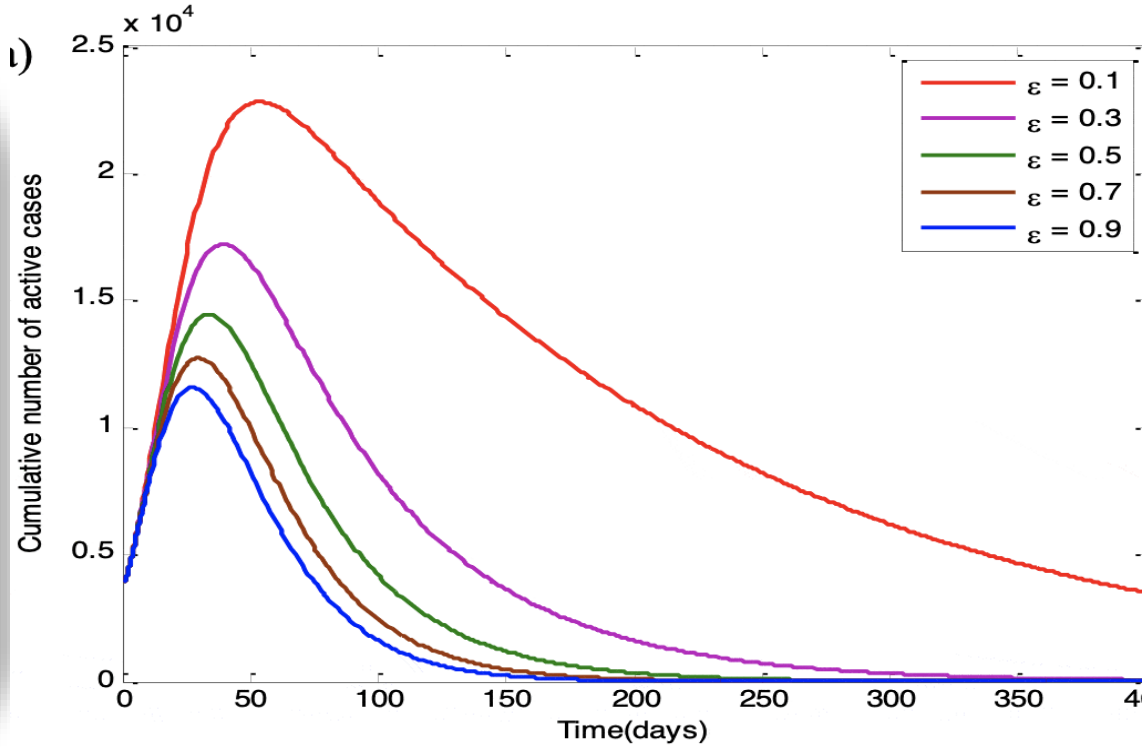
https://www.baeldung.com/wp-content/uploads/sites/4/2023/03/text.gif

Beamer: Animations - PNG

\animategraphics[<options>]{<frames per second>}{<name without extension>}{<first frame>}{<last frame>}

```
%% Beamer-Environment
\begin{frame}{GIFs in Beamer}
\centering
\animategraphics[autoplay,loop,width=4cm]{10}{fig/effects-}{0}{3}
\end{frame}
```

Output:



- Options: autoplay (the images play automatically), loop (once the images start moving, it continues in a loop (without stopping)
- frames per second]
- animation filename: must specify the animation filename without its extension and without the number
- first frame: the number of the first image
- last frame: number of the images

Beamer: Animations – GIF and Video

```
\documentclass {beamer}
\usepackage{multimedia}
\begin{document}
\movie[options]{optional text}{GIF filename}
\end{document}
```

- Options : autostart, loop, and width

Beamer: Bibliography

- Used in LaTeX environment to create unordered lists

```
\begin{frame} {References}
        \nocite{*}
        \bibliographystyle{unsrt}
        \bibliography{dvg-ref}
\end{frame}
```

```
@article{TOUBAKH2021222,
title = {Self adaptive learning scheme for
early diagnosis of simple and multiple switch
faults in multicellular power converters},
journal = {ISA Transactions},
volume = {113},
pages = {222-231},
year = {2021},
author = {Houari Toubakh and Moamar Sayed-
Mouchaweh and Mohammed Benmiloud and Michael
Defoort and Mohamed Djemai},
}
```

References

Houari Toubakh, Moamar Sayed-Mouchaweh, Mohammed Benmiloud, Michael Defoort, and Mohamed Djemai.
Self adaptive learning scheme for early diagnosis of simple and multiple switch faults in multicellular power converters.
ISA transactions, 113:222–231, 2021.

M Benmiloud and A Benalia.
Hybrid control scheme for multicellular converter.
In *2013 International Conference on Control, Decision and Information Technologies (CoDIT)*, pages 476–482. IEEE, 2013.

Mohammed Benmiloud, Atallah Benalia, Mohamed Djemai, and Michael Defoort.
Hybrid control design for limit cycle stabilisation of planar switched systems.
International Journal of Control, 91(7):1720–1729, 2018.

Beamer: Overlay - Pause

- Used in LaTeX environment to create ordered lists

```
\begin{enumerate}
\item Shown from first slide on.
\pause
\item Shown from second slide on.
\pause
\item Shown from third slide on.
\pause
\item Shown from fourth slide on.
\end{enumerate}
```

1

1. Shown from first slide on.
2. Shown from second slide on.
3. Shown from third slide on.
4. Shown from fourth slide on.

2

1. Shown from first slide on.
2. Shown from second slide on.
3. Shown from third slide on.
4. Shown from fourth slide on.

3

1. Shown from first slide on.
2. Shown from second slide on.
3. Shown from third slide on.
4. Shown from fourth slide on.

4

1. Shown from first slide on.
2. Shown from second slide on.
3. Shown from third slide on.
4. Shown from fourth slide on.

Graphics

- To include graphics in a Beamer presentation we can use the \includegraphics just like a regular LaTeX document

\includegraphics[options]{filename}

```
\documentclass{beamer}
\usepackage{graphicx}  % Package for adding figures
\begin{document}
\begin{figure}[h]
\centering
 \includegraphics{galaxy.jpg}
 \caption{Galaxy of Stars}
\end{figure}
\end{document}
```



Figure 1: Galaxy of Stars

Options are:

scale=<value>: scale the picture by <value>
height=<len>: scale the picture so that the width is <len>
width=<len>: scale the picture so that the width is <len>
angle=<x>: rotate the picture by <x> degrees
draft: Don’t display image, print filename in a box of the same size.

Thank you!

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